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TO WHOMSOEVER IT MAY CONCERN

This certificate is awarded to **Ms. Fidha Manaph** in recognition of her successful completion of the **One-Month Machine Learning Internship Program** offered by **Skoliko** on **03 June 2026.**

During the internship period, she successfully completed an **end-to-end machine learning project** titled **"Global Mobility Analyser Application"** and demonstrated **dedication, commitment, and proficiency** in applying machine learning concepts and techniques.

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Detailed Project Report

Global Mobility Application Analyzer

(An End-to-End MLOps-Based Visa Approval System)

Submitted By

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Acknowledgement

We would like to express our sincere gratitude to the entire Skoliko team for providing us with the opportunity to undertake this Machine Learning Internship and work on a real-world industry-oriented project.

This internship provided us with valuable exposure to Machine Learning, MLOps, cloud deployment, software development practices, and project documentation. Throughout the internship, we received continuous guidance, constructive feedback, and support, which helped us strengthen both our technical and professional skills.

We would also like to thank our mentors and the Skoliko team for encouraging us to explore industry-standard tools and methodologies, including machine learning pipelines, FastAPI development, AWS cloud services, Docker, CI/CD workflows, and MLOps practices. Their insights and feedback played an important role in the successful completion of this project.

Finally, we would like to thank everyone who contributed directly or indirectly to the successful completion of the project titled **"Global Mobility Application Analyzer – An End-to-End MLOps-Based Visa Approval Prediction System."**

1.

Abstract

The Global Mobility Application Analyzer is an end-to-end Machine Learning and MLOps project developed to predict whether a visa application is likely to be certified or denied based on historical immigration data. Visa approval decisions depend on multiple applicant and employer-related factors, making manual evaluation a complex and time-consuming process. The objective of this project is to automate and support the decision-making process through predictive analytics.

The project utilizes the EasyVisa dataset containing visa application records with features such as education level, work experience, prevailing wage, employer information, and employment details. Comprehensive Exploratory Data Analysis (EDA), data preprocessing, feature engineering, and data transformation techniques were applied to prepare the dataset for model training.

Multiple machine learning algorithms including Logistic Regression, Random Forest, AdaBoost, Gradient Boosting, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Classifier, XGBoost, and CatBoost were trained and evaluated. Hyperparameter tuning was performed using GridSearchCV with 3-Fold Cross-Validation, resulting in K-Nearest Neighbors emerging as the best-performing model with an accuracy of approximately 94.53%.

To ensure scalability and maintainability, the project was implemented using an end-to-end MLOps architecture comprising data ingestion, validation, transformation, model training, evaluation, and deployment pipelines. AWS S3 was used for cloud-based model storage, Evidently AI was integrated for data drift monitoring, and FastAPI was used to develop a real-time prediction web application. The deployment workflow was further automated using Docker, GitHub Actions, Amazon ECR, and AWS EC2.

The final system demonstrates how Machine Learning, MLOps, cloud technologies, and modern deployment practices can be integrated to build a scalable, production-ready visa approval prediction solution.

Contents

1. Objectives

The major objective of this project is to build a smart prediction system that estimates whether a visa application will be approved or denied using historical immigration data. The system helps immigration consultants, employers, and applicants understand approval probability and identify important risk factors affecting visa decisions.

Main Objectives

- Build ML classification models for visa approval prediction
- Predict approval status using historical immigration data
- Identify important factors affecting visa approval
- Automate application triage and risk assessment
- Create end-to-end ML pipeline and deployment system
- Develop a web application for real-time prediction

2. Benefits

- Helps applicants improve profiles before applying
- Assists immigration consultants in decision support
- Reduces manual screening effort
- Enables faster risk assessment of applications
- Helps employers estimate sponsorship success rate
- Supports automated and data-driven decision making
- Allows monitoring of model quality using data drift detection
- Improves scalability through deployment and cloud integration

3. The Problem Statement

Visa approval processes involve many factors such as education, experience, wage, company profile, and region of employment. Manual analysis of applications is time-consuming and may lead to inconsistent decisions.

The aim of this project is to build a predictive machine learning model that determines whether a visa application is likely to be approved or denied based on applicant and employer information.

The system uses historical immigration application data containing approved and denied cases.

4. Frame the problem

Before working with the immigration dataset, it is important to understand how the organization expects to use and benefit from the prediction model. This initial analysis helps determine the type of machine learning approach, the suitable algorithms, and the performance evaluation methods required for the project.

a) Type of Machine Learning Problem

The Global Mobility Application Analyzer is a **Supervised Machine Learning problem** because the dataset contains labeled historical visa application records where the approval status is already known.

The model learns patterns from previously approved and denied applications and uses them to predict outcomes for new applicants.

b) Nature of the Prediction Task

This project is a **Classification problem** because the model predicts categorical outcomes:

- Approved
- Denied

The target variable is binary:

- 1 = Approved
- 0 = Denied

Future extensions may also include multiclass prediction categories such as:

- Pending
- Withdrawn

c) Learning Approach

The project mainly follows a **Batch/Offline Learning approach** because the model is trained using historical immigration datasets collected over a period of time.

There is no continuous real-time stream of data requiring instant retraining. The dataset is manageable in size and periodic retraining can be performed whenever updated immigration records become available.

The system may later incorporate monitoring and retraining workflows if significant data drift is detected.

d) Performance Measurement

Since the project involves classification, model performance is evaluated using classification metrics such as:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score

The primary evaluation metric is **ROC-AUC Score**, which measures how effectively the model distinguishes between approved and denied visa applications.

Precision and Recall are also important because:

- High Recall helps identify potentially risky cases correctly.
- High Precision reduces incorrect approval predictions.

The project success target is achieving a ROC-AUC score of 0.80 or higher on unseen test data.

5. The Hypotheses

After understanding the problem statement and selecting suitable evaluation metrics, it is useful to form assumptions regarding factors that may influence visa approval outcomes based on the available immigration dataset.

These hypotheses help guide exploratory data analysis, feature engineering, and model development.

5.1 Applicant Level Hypotheses

Education

Applicants with higher educational qualifications such as Master's Degree or Doctorate are expected to have higher visa approval probability because they possess advanced skills and qualifications.

Job Experience

Applicants with previous work experience are likely to have higher approval chances because experienced professionals are generally more employable.

Required Training

Applicants requiring less additional training may receive higher approval rates because employers often prefer candidates who can contribute immediately.

Continent

Approval rates may vary across continents due to immigration policies, labor demand, and regional applicant trends.

Prevailing Wage

Applications with higher prevailing wages are expected to have higher approval probability because they indicate better employment standards and financial stability.

Full-Time Employment

Full-time employment positions are more likely to receive approval compared to part-time roles because they indicate stable employment conditions.

5.2 Employer Level Hypotheses

Company Size

Large organizations with more employees are expected to have higher visa approval rates because they are considered financially stable and operationally reliable.

Age of Company

Older companies may achieve better approval outcomes because long-established organizations generally have stronger credibility and sponsorship history.

Region of Employment

Regions with higher workforce demand may show higher approval rates because of labor shortages and economic requirements.

6. Data Exploration

The EasyVisa dataset contains 25,460 observations and consists of multiple applicant and employer-related attributes collected from historical visa application records. The dataset includes features such as continent, education of employee, job experience status, job training requirement, number of employees, year of establishment, region of employment, prevailing wage, wage unit, and full-time position status.

Out of these attributes, the response variable (target variable) is the case_status attribute, which represents whether the visa application was approved or denied, while the remaining attributes are used as predictor variables for model training. Most of the features in the dataset fall under applicant-level and employer-level categories, where applicant-level attributes include education, work experience, and training requirements, while employer-level attributes include company size, establishment year, wage details, and employment region.

After preprocessing and feature engineering, the dataset was divided into training and testing sets in the ratio 80:20 for model training and evaluation.

7. Strategy: Solving the Problem with Machine Learning and MLOps

1. Automated Visa Approval Prediction
The system leverages machine learning models trained on historical visa application data to predict whether a visa application is likely to be Certified or Denied based on applicant and employer-related attributes.

2. Reducing Manual Screening Effort
By automating the prediction process, the system assists in reducing the effort required for manually reviewing large volumes of visa applications while improving consistency in decision-making.

3. Data-Driven Decision Support
The model identifies patterns from historical visa records and uses them to generate prediction outcomes, enabling faster and more informed decision support.

4. Reliable Machine Learning Pipeline

The project incorporates automated data validation, schema verification, preprocessing, transformation, model training, evaluation, and deployment pipelines to ensure reliable and reproducible results.

5. Continuous Data Quality Monitoring
The FastAPI application is containerized using Docker and deployed on AWS cloud infrastructure to provide scalable and real-time prediction services.

6. Scalable Cloud-Based Deployment

The trained model is deployed through a FastAPI web application and integrated with AWS S3 cloud storage, enabling scalable, centralized, and real-time prediction services.

8. Methodology and Analysis Workflow

The objective of this project is to develop a machine learning system capable of predicting visa approval outcomes while implementing industry-standard MLOps practices. The project workflow is organized into the following stages:

1. Data Collection and Ingestion

Historical visa application records are collected from MongoDB and converted into structured Pandas DataFrames. The data is stored in a feature store and divided into training and testing datasets.

2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand feature distributions, identify missing values, detect class imbalance, analyze relationships between variables, and uncover patterns affecting visa approval decisions.

3. Data Validation and Drift Detection

The dataset is validated against a predefined schema configuration to verify the presence of required numerical and categorical features. Evidently AI is used to detect data drift and ensure dataset consistency.

4. Data Preprocessing and Feature Engineering

Missing values are handled using appropriate imputation techniques, duplicate records are removed, categorical features are encoded, and numerical features are scaled. A new feature called company_age is created, and skewed numerical variables are transformed using the Yeo-Johnson Power Transformation.

5. Data Transformation

A preprocessing pipeline is built using Scikit-learn's Pipeline and ColumnTransformer to automate feature transformation and maintain consistency between training and prediction workflows.

6. Model Training

Multiple machine learning classification algorithms were trained and evaluated to identify the most suitable model for visa approval prediction. The models trained include:

- Logistic Regression
- Random Forest Classifier
- AdaBoost Classifier
- Gradient Boosting Classifier
- Decision Tree Classifier
- K-Nearest Neighbors (KNN)
- Support Vector Classifier (SVC)
- XGBoost Classifier
- CatBoost Classifier

Each model was trained using the transformed dataset and evaluated using multiple performance metrics, including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The objective was to compare model performance and identify the best candidate for hyperparameter tuning and deployment.

7. Hyperparameter Tuning and Model Selection

After the initial model comparison, the top-performing models were further optimized using **GridSearchCV with 3-Fold Cross-Validation**. Hyperparameter tuning was performed to identify the optimal parameter combinations and improve predictive performance.

Following tuning and evaluation, **K-Nearest Neighbors (KNN)** emerged as the best-performing model, achieving approximately **94.53% accuracy**, and was selected as the final production model for deployment.

8. Model Evaluation

The newly trained model is compared against the existing production model stored in AWS S3. Only models demonstrating improved performance are approved for deployment.

9. Model Deployment and Cloud Integration

The finalized model is uploaded to AWS S3 cloud storage using the Model Pusher component. AWS S3 serves as a centralized repository for model artifacts and deployment resources.

10. FastAPI Application and Prediction Pipeline

A FastAPI-based web application is used to collect user input, execute the prediction pipeline, and display real-time visa approval predictions. The same preprocessing pipeline used during training is applied before generating predictions.

11. Continuous Deployment

The application is containerized using Docker and deployed using GitHub Actions, Amazon ECR, and AWS EC2, enabling automated CI/CD workflows and scalable deployment.

9. Prediction

Prediction involves estimating whether a visa application is likely to be approved or denied based on historical immigration application data. Machine learning algorithms improve the system’s ability to identify hidden patterns and relationships between applicant and employer-related features without manual intervention. In this project, machine learning classification algorithms such as K-Nearest Neighbors (KNN) and Random Forest (RF) are used for visa approval prediction.

9.1. K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a supervised machine learning classification algorithm used for predicting the class of a data point based on its nearest neighboring data samples. The algorithm identifies the ‘K’ closest data points to the given input using distance measures such as Euclidean distance and assigns the majority class among the neighbors as the final prediction.

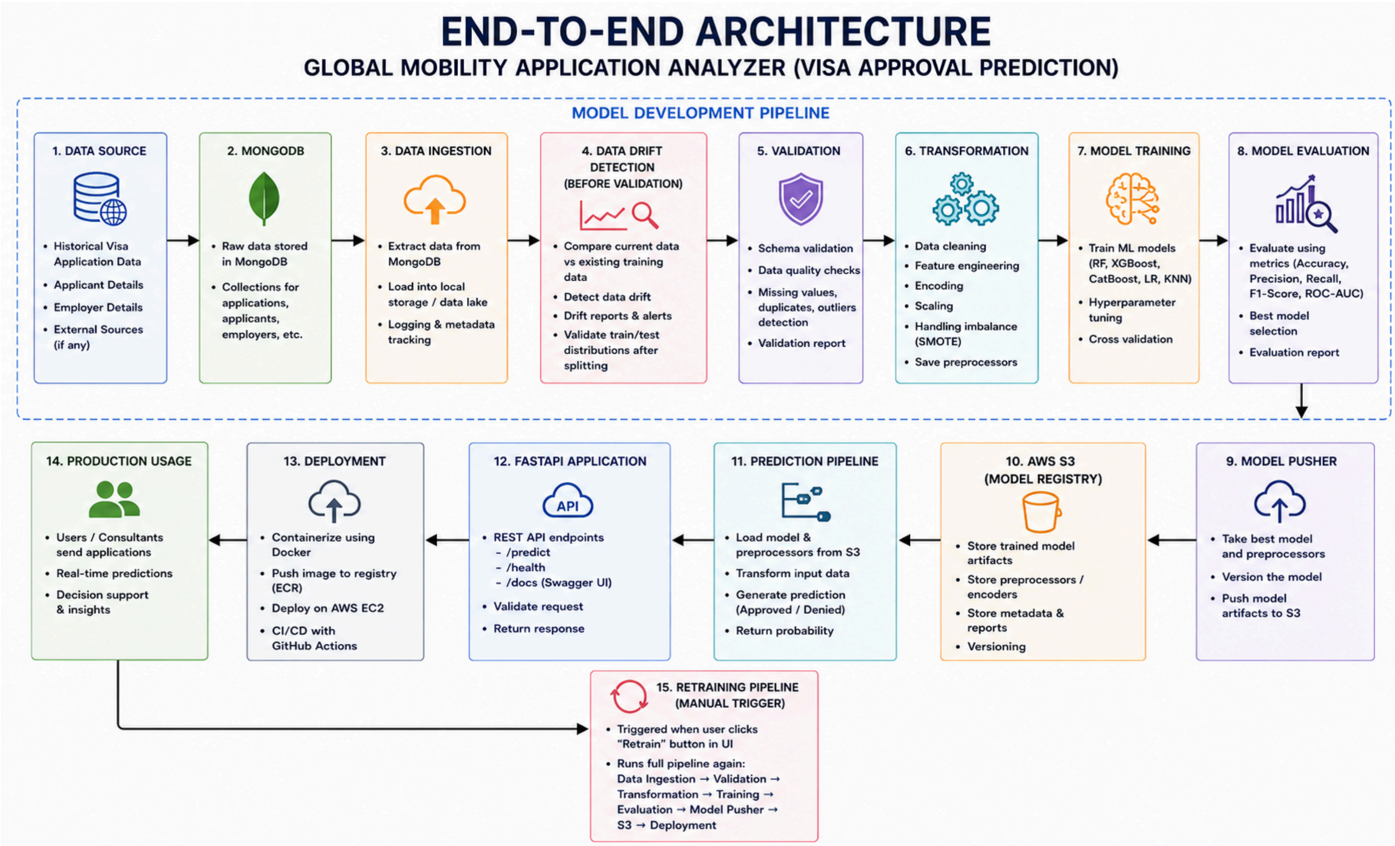
In this project, KNN predicts visa approval or denial by comparing a new visa application with similar historical applications based on features such as education level, work experience, prevailing wage, employer size, and employment region. The prediction outcome depends on the similarity between the new application and previously observed cases.

9.2. Random Forest

Random Forest is a tree-based ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each decision tree is trained using randomly selected subsets of the dataset and features, and the final prediction is determined through majority voting among all trees.

In this project, Random Forest is used to analyze complex relationships between applicant and employer-related attributes for predicting visa approval outcomes. The algorithm is capable of handling both numerical and categorical features effectively and also helps identify important features influencing visa approval decisions through feature importance analysis.

10. System Architecture



The Global Mobility Application Analyzer follows an end-to-end MLOps architecture that automates the complete machine learning lifecycle.

The workflow begins with historical visa application records stored in MongoDB. Data is extracted through the Data Ingestion component and passed through Data Validation, Drift Detection, Data Preprocessing, Feature Engineering, and Data Transformation stages.

The transformed dataset is then used for Model Training and Hyperparameter Tuning. The best-performing model is evaluated and approved through the Model Evaluation component before being pushed to AWS S3 cloud storage using the Model Pusher component.

For real-time prediction, FastAPI loads the latest production model from AWS S3. User inputs are processed through the Prediction Pipeline and passed to the trained model to generate visa approval predictions.

The architecture supports automation, scalability, monitoring, and future model retraining workflows.

11. Technology Stack

Programming Language
• Python

Data Processing
• Pandas
• NumPy

Machine Learning
• Scikit-learn
• XGBoost
• CatBoost

Database
• MongoDB
• PyMongo

MLOps & Monitoring
• Evidently AI

Web Framework
• FastAPI

Cloud Services
• AWS S3
• Amazon EC2
• Amazon ECR

Containerization
• Docker

Version Control & CI/CD
• Git
• GitHub
• GitHub Actions

Development Environment
• Jupyter Notebook
• VS Code

12. Model Performance Summary

Multiple machine learning algorithms were trained and evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics.

The evaluated models include:

• Logistic Regression
• Random Forest Classifier
• AdaBoost Classifier
• Gradient Boosting Classifier
• Decision Tree Classifier
• K-Nearest Neighbors (KNN)
• Support Vector Classifier (SVC)
• XGBoost Classifier
• CatBoost Classifier

Before hyperparameter tuning, Random Forest achieved the best performance with approximately 91.1% accuracy.

After applying GridSearchCV with 3-Fold Cross-Validation, K-Nearest Neighbors (KNN) emerged as the best-performing model with an accuracy of approximately 94.53% and was selected as the final production model.

13. Conclusion

The Global Mobility Application Analyzer successfully demonstrates the implementation of an end-to-end machine learning and MLOps solution for visa approval prediction.

The project integrates data ingestion, validation, transformation, model training, evaluation, cloud storage, deployment, and real-time prediction into a unified workflow. Through this system, historical immigration data is transformed into actionable insights that support faster and more consistent decision-making.

The project also provided practical exposure to Machine Learning Engineering, MLOps practices, FastAPI development, cloud technologies, CI/CD automation, and production-level system design.

14. Future Scope

• Integrate MLflow for experiment tracking and model versioning.

• Deploy the application using Kubernetes for improved scalability and reliability.

• Implement user authentication and role-based access control.

• Develop real-time monitoring dashboards for model performance and system health.

• Extend the system to support additional visa categories and multiclass prediction outcomes.

References

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2. Skoliko